Fusion reactions

Christopher Albert

Main fusion reactions

D–T	$\mathrm{D} + \mathrm{T} \rightarrow {}^{4}\mathrm{He}\ (3.5\ \mathrm{MeV}) + \mathrm{n}\ (14.1\ \mathrm{MeV})$
D–D	$\mathrm{D} + \mathrm{D} \rightarrow \mathrm{T}\ (1.01\ \mathrm{MeV}) + \mathrm{p}\ (3.02\ \mathrm{MeV})$
	$\mathrm{D} + \mathrm{D} \rightarrow {}^{3}\mathrm{He}\ (0.82\ \mathrm{MeV}) + \mathrm{n}\ (2.45\ \mathrm{MeV})$
D–^{3}He	$\mathrm{D} + {}^{3}\mathrm{He} \rightarrow {}^{4}\mathrm{He}\ (3.6\ \mathrm{MeV}) + \mathrm{p}\ (14.7\ \mathrm{MeV})$
p–^{11}B	$\mathrm{p} + {}^{11}\mathrm{B} \rightarrow 3\,{}^{4}\mathrm{He} \quad (8.7\ \mathrm{MeV})$
p–p (Sun)	$\mathrm{p} + \mathrm{p} \rightarrow \mathrm{D} + \mathrm{e}^{+} + \nu_e \quad (0.42\ \mathrm{MeV})$

Two-body Coulomb problem

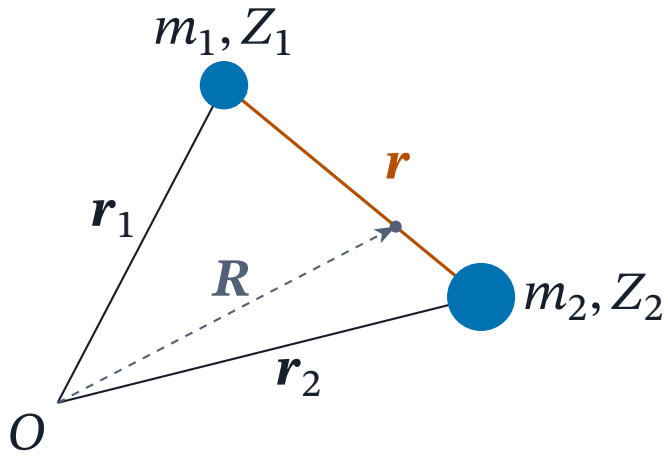
$$H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 \, |\boldsymbol{r}_1 - \boldsymbol{r}_2|}$$

$$M = m_1 + m_2, \quad m = \frac{m_1 m_2}{m_1 + m_2}$$

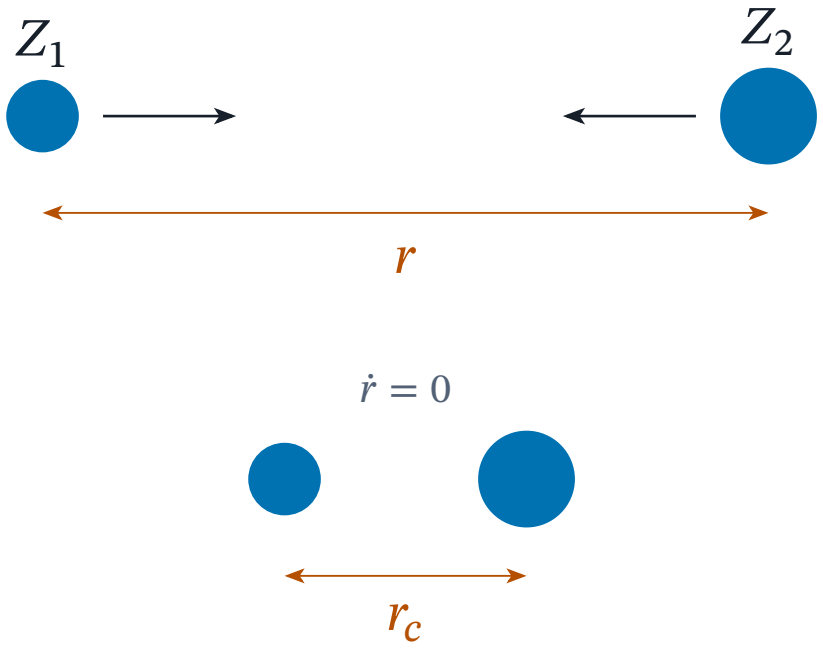
$$\boldsymbol{R} = \frac{m_1 \boldsymbol{r}_1 + m_2 \boldsymbol{r}_2}{M}, \quad \boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$$

$$\boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2, \quad \boldsymbol{p} = \frac{m_2 \boldsymbol{p}_1 - m_1 \boldsymbol{p}_2}{M}$$

$$H = \frac{P^2}{2M} + \frac{p^2}{2m} + \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 r}$$



Head-on collision



$$\dot{\boldsymbol{P}} = -\frac{\partial H}{\partial \boldsymbol{R}} = 0 \quad \Rightarrow \quad \boldsymbol{P} = 0$$

centre-of-mass frame; head-on: $\ell = 0$

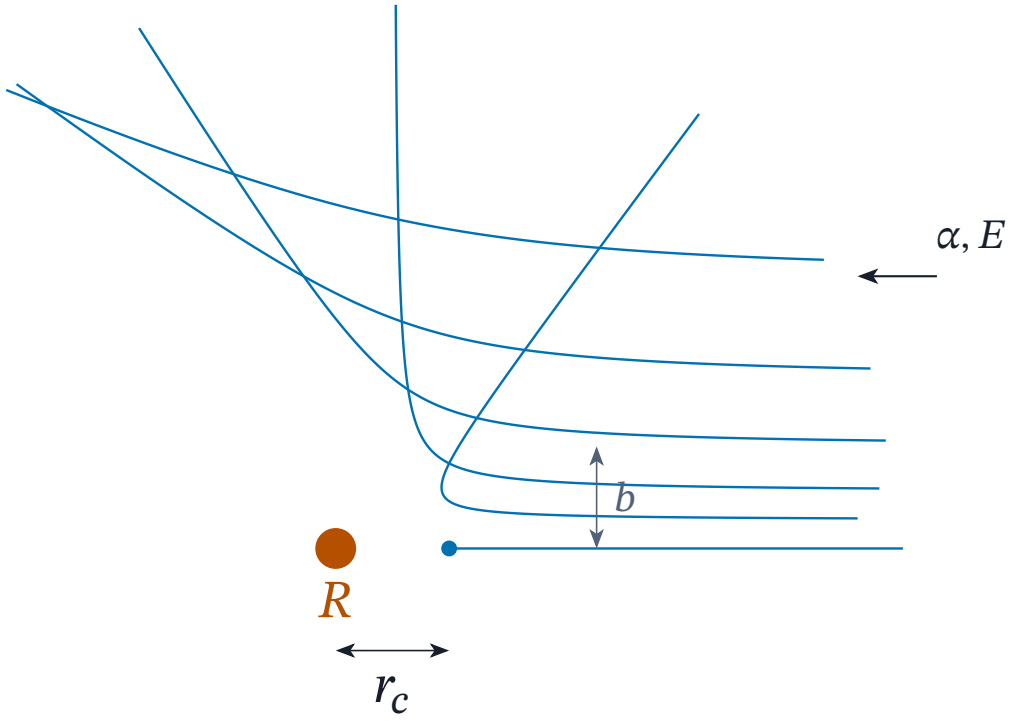
$$E = \frac{m}{2}\dot{r}^2 + V(r) = \text{const}$$

$$V(r) = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 r}$$

$$V(r_c) = E \quad \Rightarrow \quad r_c = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 E}$$

$$|t| = \sqrt{\frac{m}{2}} \int_{r_c}^{r} \frac{\mathrm{d}r'}{\sqrt{E - V(r')}}$$

Rutherford scattering



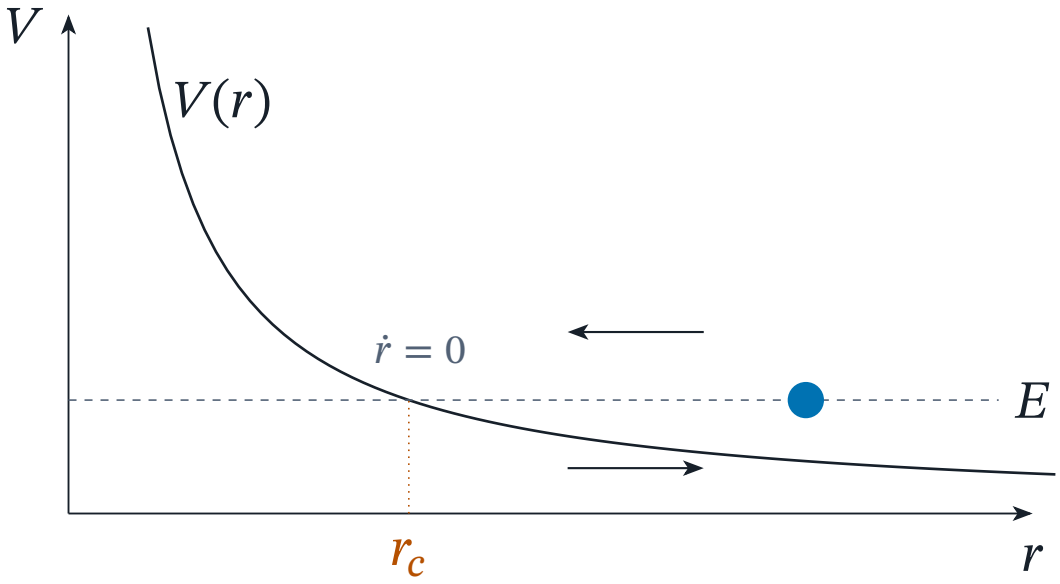
$$r_c = \frac{Z_\alpha Z e^2}{4\pi\varepsilon_0 E}$$

	Z	E	r_c	
Au	79	5 MeV	45 fm	Coulomb
Al	13	7.7 MeV	5 fm	deviation

$$r_c \approx R \quad \Rightarrow \quad R \sim 10^{-15}\ \text{m}$$

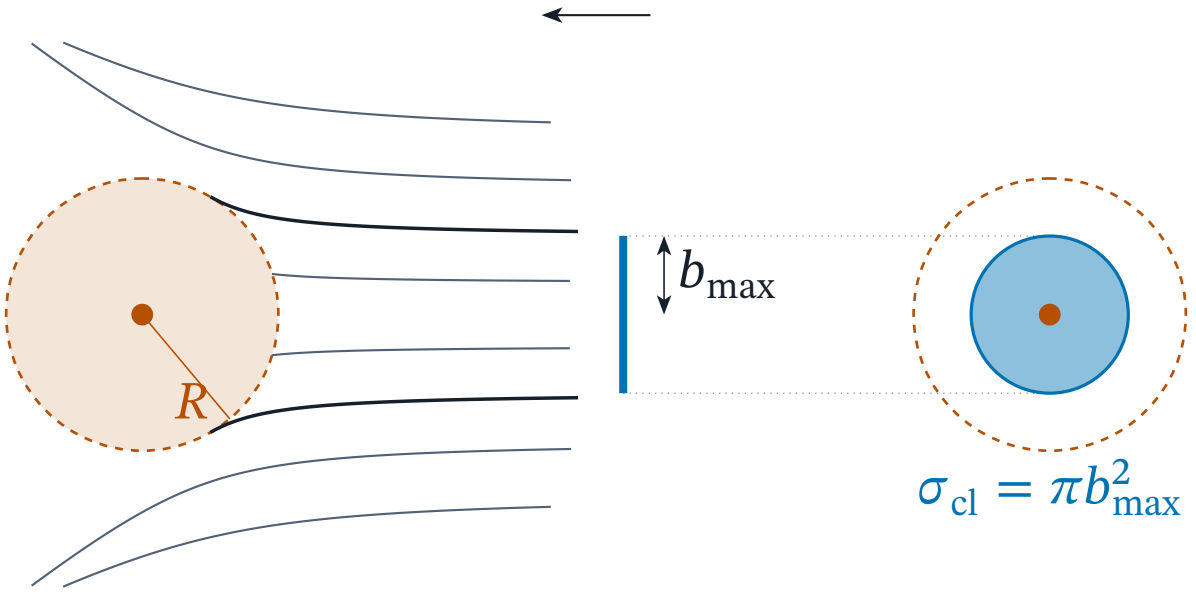
Geiger & Marsden · Proc. R. Soc. A 82, 495 (1909) · Rutherford · Phil. Mag. 21, 669 (1911)
Bieler · Proc. R. Soc. A 105, 434 (1924) · Rutherford & Chadwick · Phil. Mag. 50, 889 (1925)

Classical turning point



$$r_c \lesssim R \approx 3\ \text{fm} \quad \Rightarrow \quad E \gtrsim \frac{e^2}{4\pi\varepsilon_0 R} \approx 0.5\ \text{MeV}$$

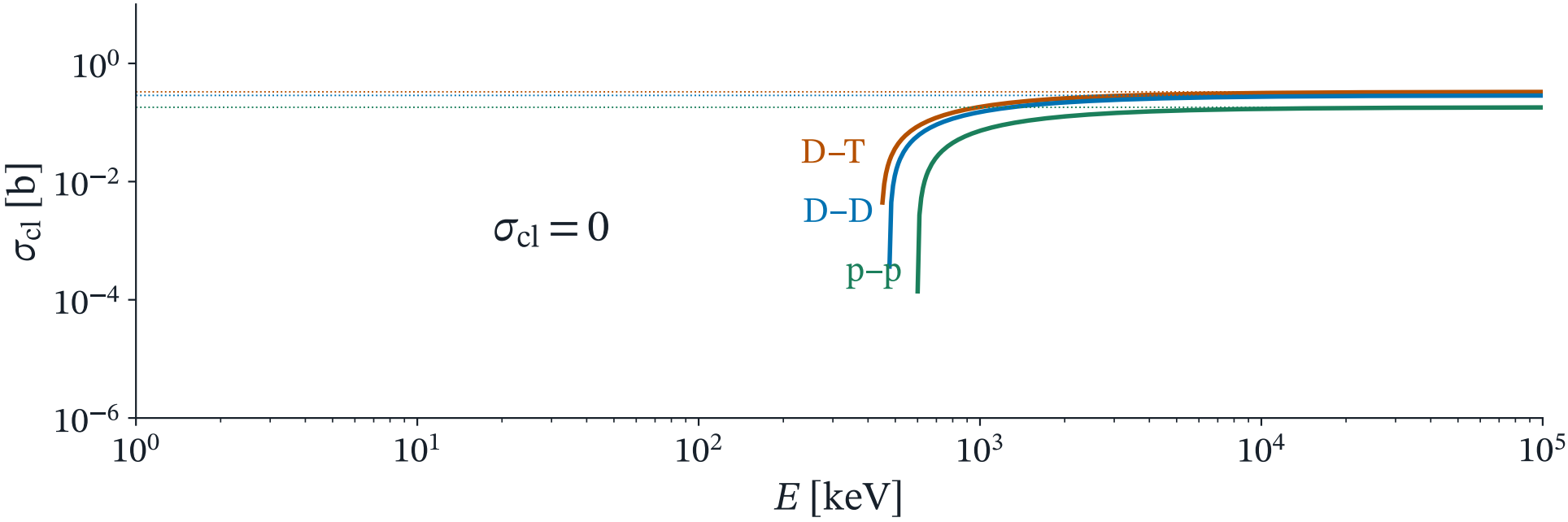
Classical cross section



$$r_{\min}(b) \le R \quad \Rightarrow \quad b^2 \le R^2\left(1 - \frac{V(R)}{E}\right)$$

$$\sigma_{\mathrm{cl}} = \pi R^2\left(1 - \frac{V(R)}{E}\right) \qquad \sigma_{\mathrm{cl}} = 0 \quad (E < V(R))$$

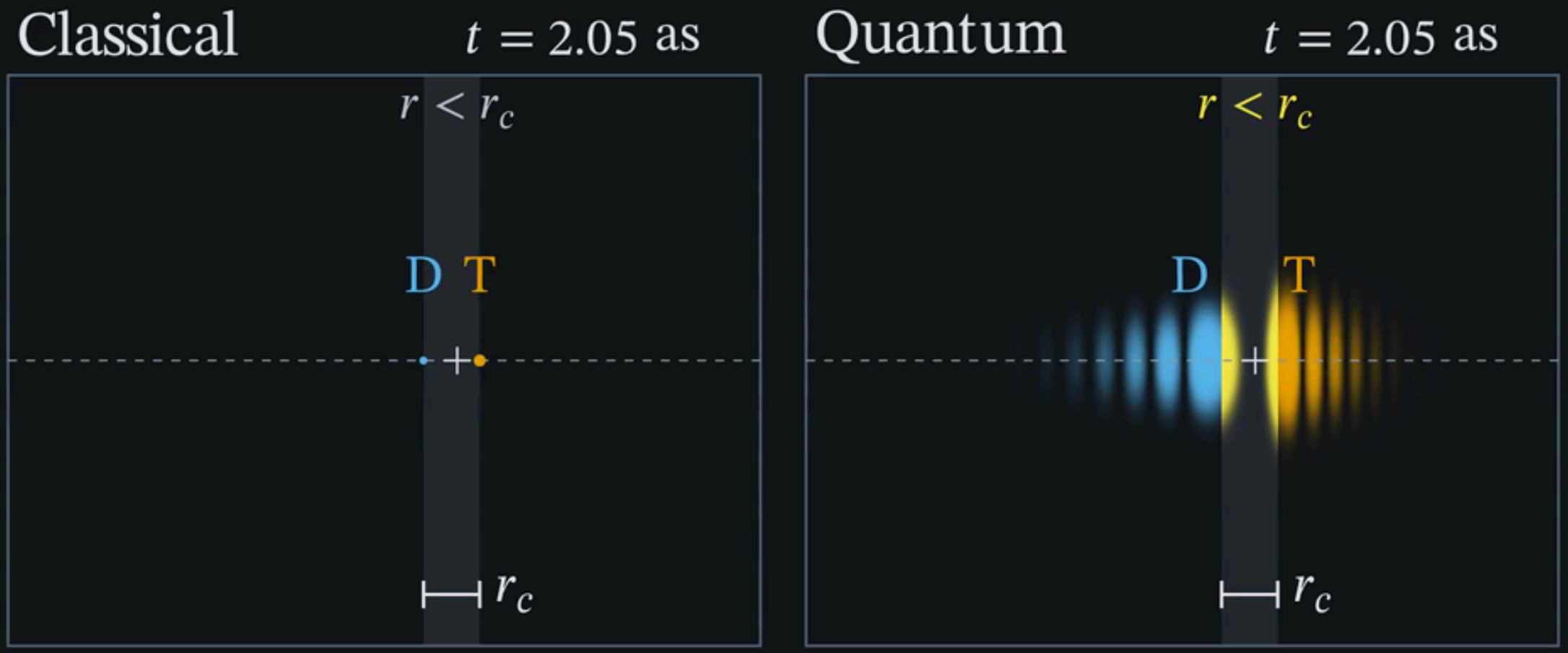
Classical cross sections



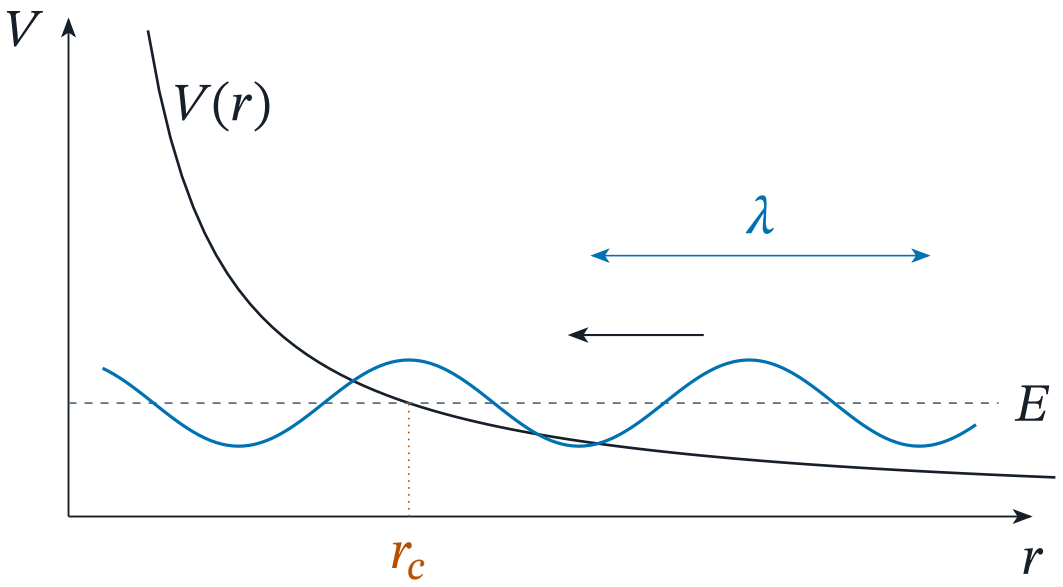
$$\sigma_{\mathrm{cl}} = \pi R^2 \left(1 - \frac{V(R)}{E}\right)$$

$$R = r_0\left(A_1^{1/3} + A_2^{1/3}\right), \quad r_0 \approx 1.2\ \mathrm{fm}$$

	A_1, A_2	R	$V(R)$	πR^2
D–T	2, 3	3.2 fm	440 keV	0.33 b
D–D	2, 2	3.0 fm	480 keV	0.29 b
p–p	1, 1	2.4 fm	600 keV	0.18 b


Classical
$t = 2.05$ as
$r < r_c$
D T
r_c
Quantum
$t = 2.05$ as
$r < r_c$
D T
r_c

Quantum ballpark



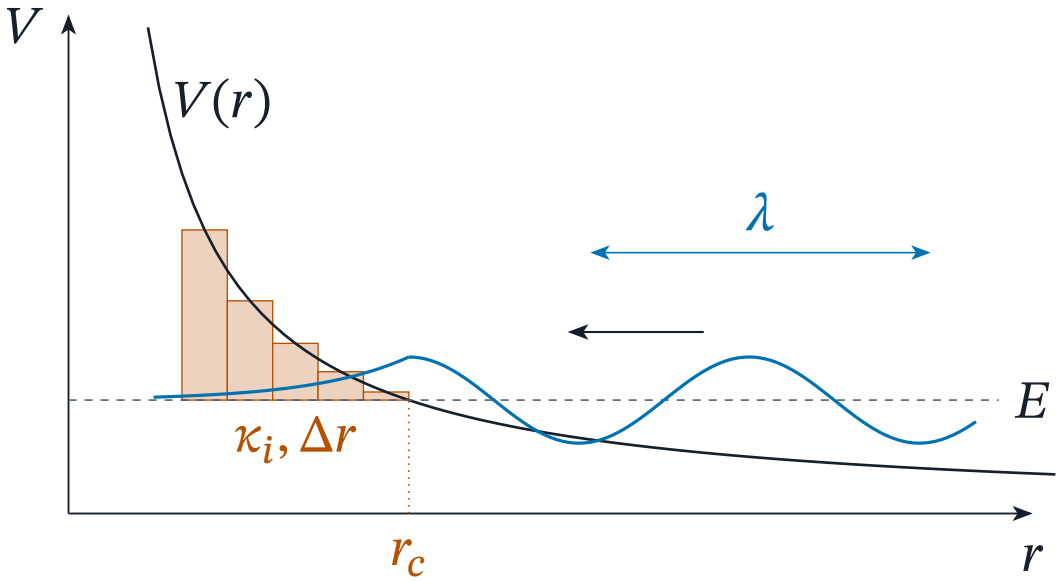
$$\lambda\!\!\!^{-} = \frac{\lambda}{2\pi} = \frac{\hbar}{p} = \frac{\hbar}{\sqrt{2mE}}$$

$$\lambda\!\!\!^{-} \approx r_c \quad \Rightarrow \quad E \approx 2mc^2(\alpha Z_1 Z_2)^2$$

D–T: $E \approx 120$ keV

$$\frac{r_c}{\lambda\!\!\!^{-}} = \alpha Z_1 Z_2 c\sqrt{\frac{2m}{E}} \propto E^{-1/2}$$

Barrier in slices



one slice:

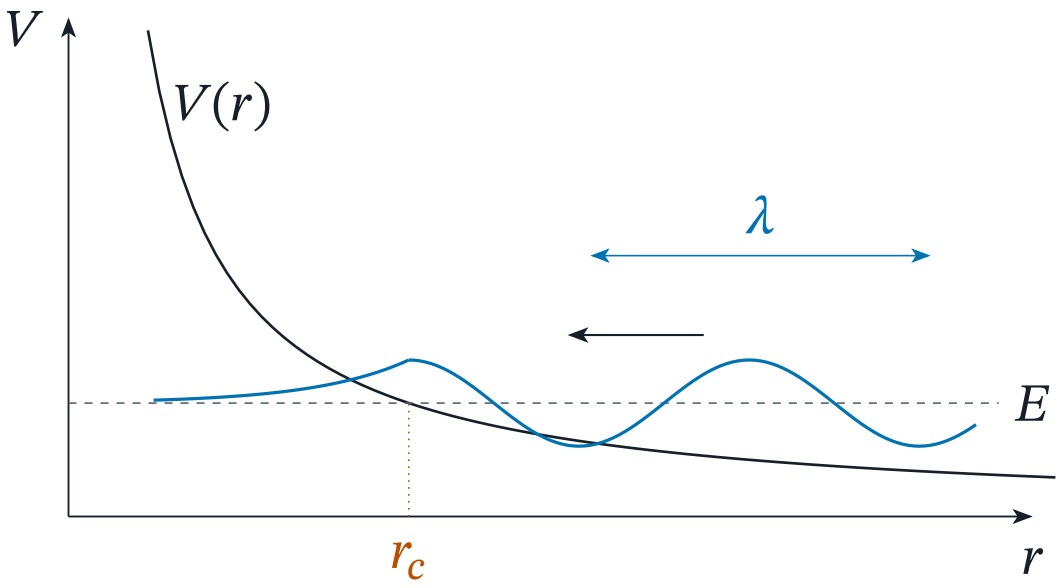
$$P_i \approx e^{-2\kappa_i \Delta r}, \quad \kappa_i = \frac{\sqrt{2m(V(r_i) - E)}}{\hbar}$$

all slices:

$$P \approx \prod_i P_i = \exp\left(-2 \sum_i \kappa_i \Delta r\right)$$

$$\to \exp\left(-2 \int_0^{r_c} \kappa(r)\, \mathrm{d}r\right)$$

Gamow tunnelling



$$V(r) = E\frac{r_c}{r} \quad \Rightarrow \quad \kappa(r) = \frac{1}{\lambda\!\!\!^{-}}\sqrt{\frac{r_c}{r} - 1}$$

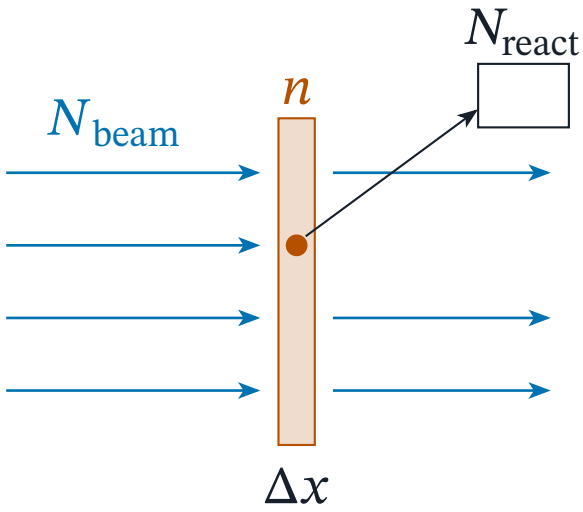
$$2\int_0^{r_c} \kappa \, \mathrm{d}r = \frac{2r_c}{\lambda\!\!\!^{-}} \int_0^1 \sqrt{\frac{1}{x} - 1}\, \mathrm{d}x = \pi\frac{r_c}{\lambda\!\!\!^{-}}$$

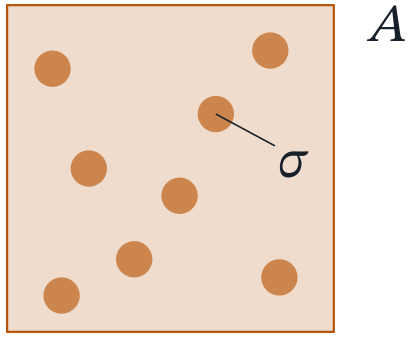
$$P \approx \exp\left(-\pi\frac{r_c}{\lambda\!\!\!^{-}}\right) = \exp\left(-\sqrt{\frac{E_G}{E}}\right)$$

$$E_G = 2mc^2(\pi\alpha Z_1 Z_2)^2$$

Gamow · Z. Phys. 51, 204 (1928) · Gurney & Condon · Nature 122, 439 (1928)
Atkinson & Houtermans · Z. Phys. 54, 656 (1929)

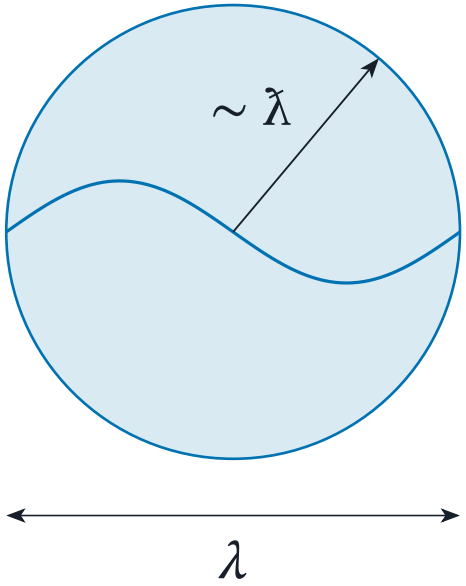
Measuring a cross section





nuclei	$N_T = nA\Delta x$
covered fraction	$P = \frac{N_T \sigma}{A} = (n\Delta x)\,\sigma$
reactions	$N_{\text{react}} = N_{\text{beam}}\, P$
measured	$\sigma = \frac{N_{\text{react}}}{N_{\text{beam}}} \frac{1}{n\Delta x}$
unit	$1\ \text{b} = 10^{-28}\text{m}^2$

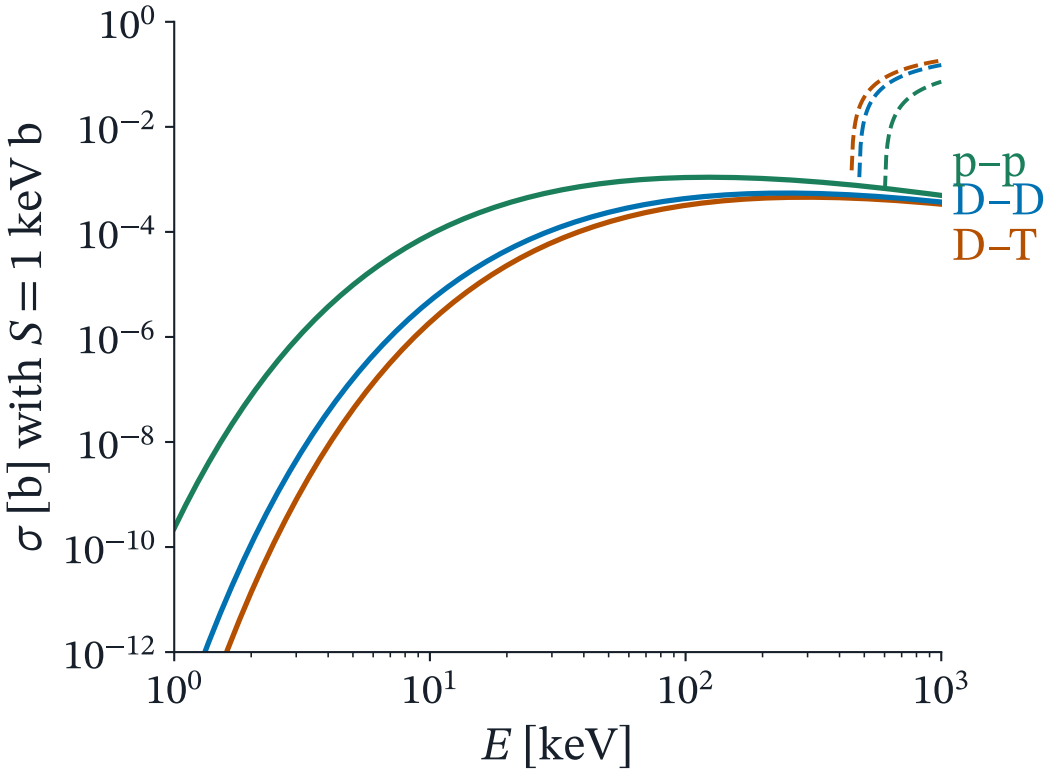
Cross section



$$\sigma_{\text{geo}} \sim \pi \lambda\!\!\!^{-2} = \frac{\pi \hbar^2}{2mE} \propto \frac{1}{E}$$

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\sqrt{\frac{E_G}{E}}\right)$$

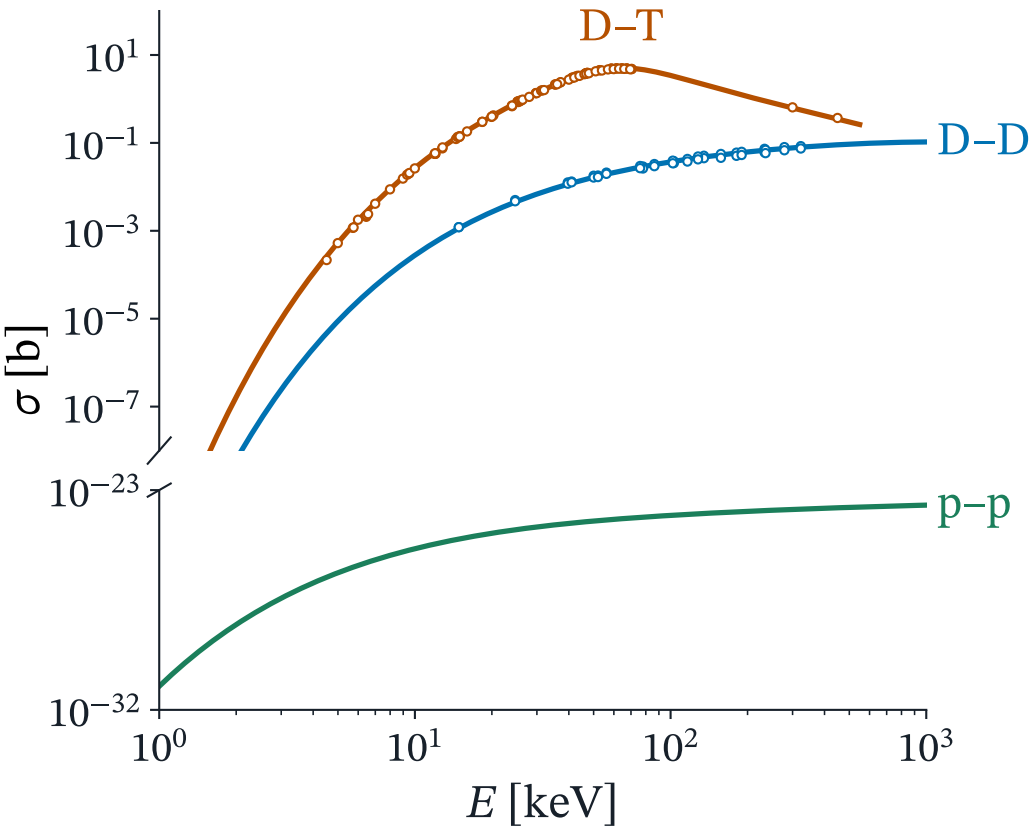
Tunnelling alone



$$\sigma = \frac{1}{E}\exp\left(-\sqrt{\frac{E_G}{E}}\right), \quad S = 1\ \text{keV b}$$

	E_G
D–T	1.18 MeV
D–D	0.99 MeV
p–p	0.49 MeV

Actual cross sections

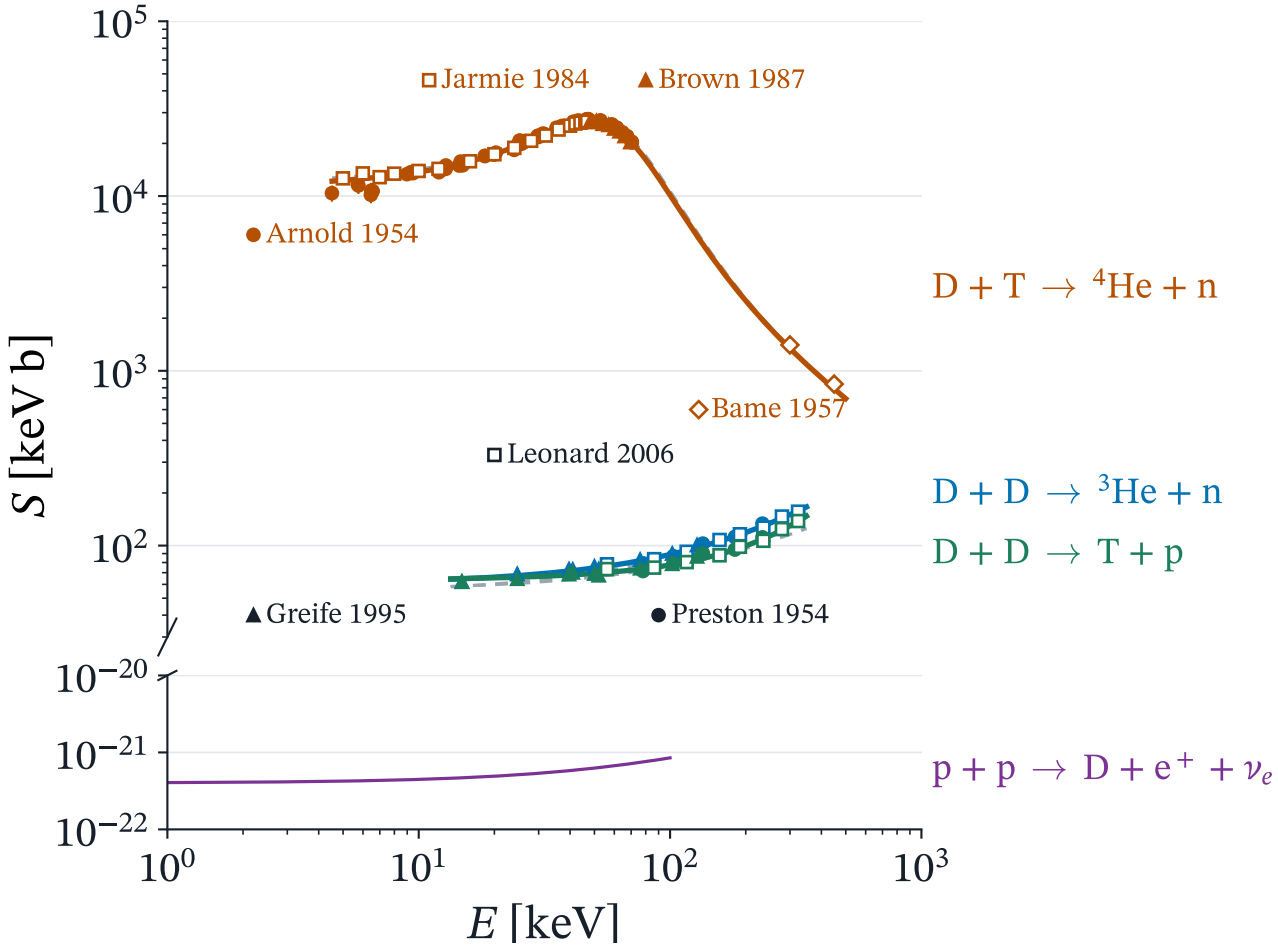


$$\sigma = \frac{S(E)}{E} \exp\left(-\sqrt{\frac{E_G}{E}}\right)$$

	$S(0)$
D–T	$\approx 10^{4}$ keV b
D–D	$\approx 10^{2}$ keV b
p–p	4×10^{-22} keV b

Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)

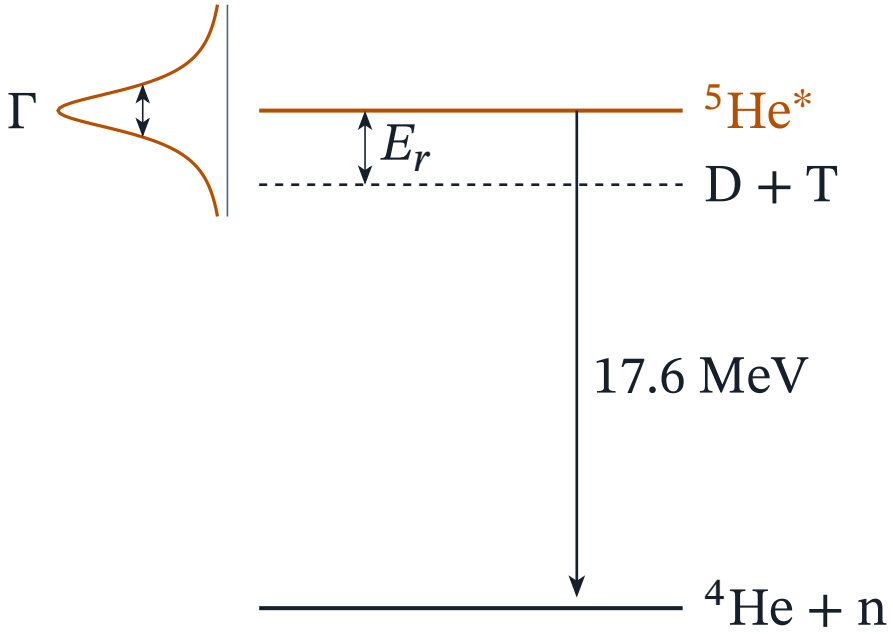
S-factor from experiment



$$S_i = \sigma_i E_i \exp\left(\sqrt{\frac{E_G}{E_i}}\right)$$

Arnold · Phys. Rev. 93, 483 (1954) · Bame · Phys. Rev. 107, 1616 (1957) · Jarmie · Phys. Rev. C 29, 2031 (1984) · Brown · Phys. Rev. C 35, 1999 (1987) · Preston · Proc. R. Soc. A 226, 206 (1954) · Greife · Z. Phys. A 351, 107 (1995) · Leonard · Phys. Rev. C 73, 045801 (2006) · Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)

^{5}He resonance



$$S(E) \propto \frac{1}{(E - E_r)^2 + \Gamma^2/4}$$

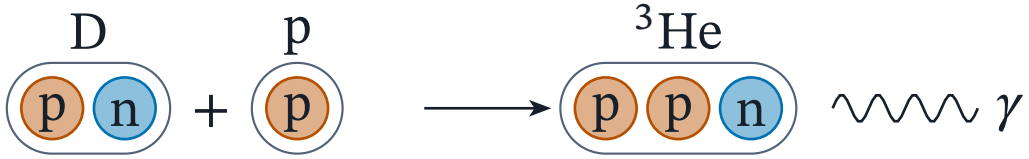
$$\tau = \frac{\hbar}{\Gamma}$$

Which force?

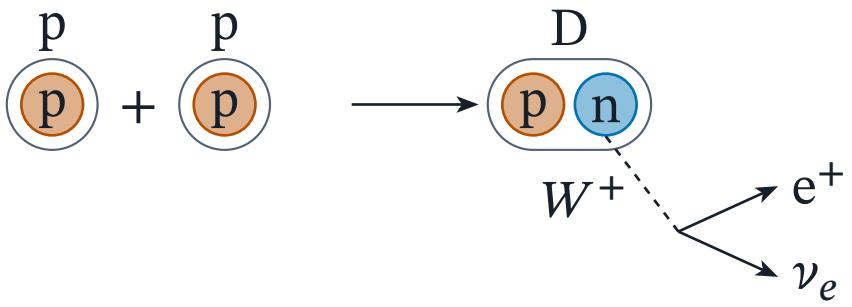


strong
$P \sim 1$

also D–D, D–^{3}He, p–^{11}B



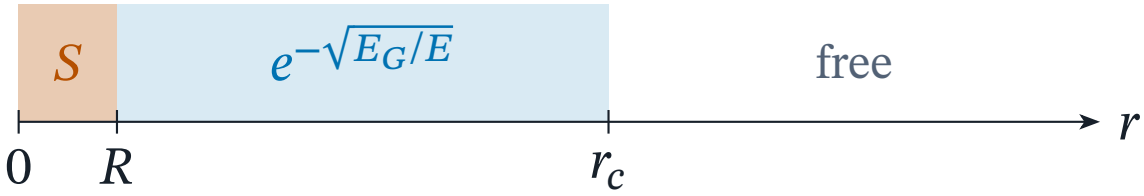
electromagnetic
$P \sim 10^{-5}$



weak
$P \sim 10^{-22}$

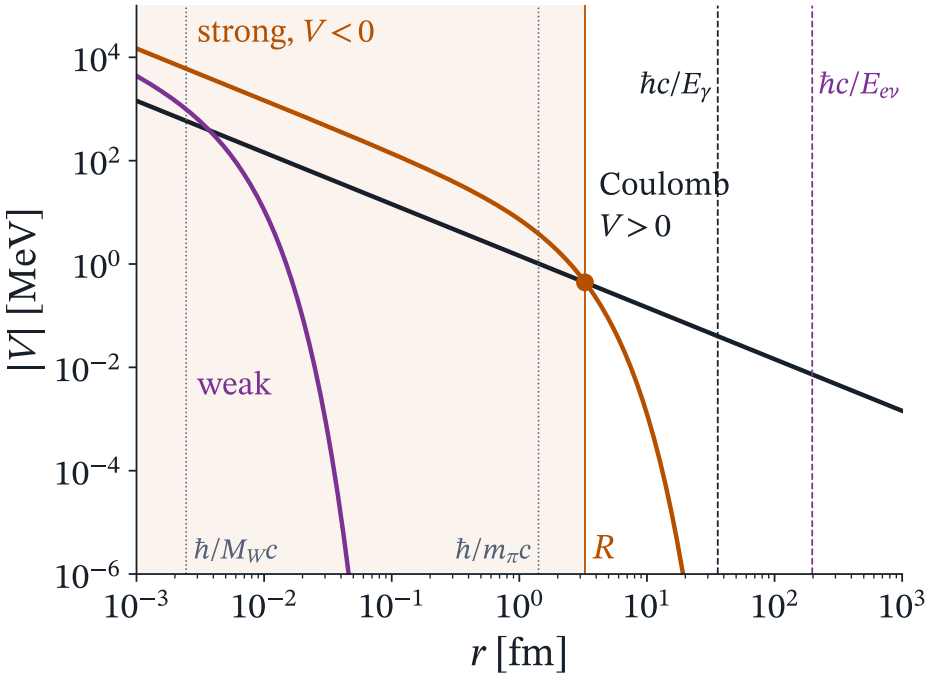
Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)
CODATA 2022 · Rev. Mod. Phys. 97, 025002 (2025)

S and R



$$\sigma = \frac{S}{E} e^{-\sqrt{E_G/E}}$$

Range of the forces



$$\lambda = \frac{\hbar}{mc}$$

Fermi's golden rule

$$P \propto \left| \int \psi_f^* \, V \, \psi_i \mathrm{d}^3 r \right|^2$$

$$\sim \begin{cases} 1 & \text{strong} \\ \alpha \left(R/\lambda_\gamma\right)^2 \approx 10^{-5} & \text{electromagnetic} \\ \alpha_W^2 (\lambda_W/\lambda_{e\nu})^4 \approx 10^{-22} & \text{weak} \end{cases}$$

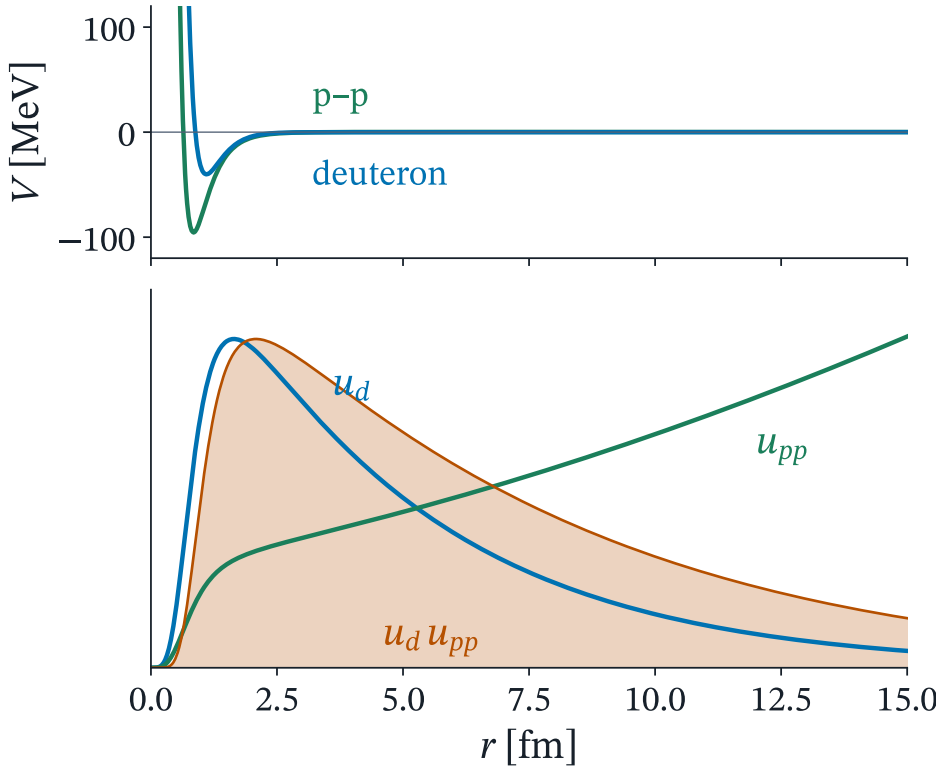
Computing S

$$S \propto e^{4\sqrt{R/a}} \cdot P \qquad a = \frac{4\pi\varepsilon_0\hbar^2}{2mZ_1Z_2e^2}$$

p–p	weak interaction	$S \propto G_F^2 g_A^2 \lvert\langle \mathrm{d} \mid \mathrm{pp}\rangle\rvert^2$
D–T	R-matrix	E_r, Γ fitted $\Rightarrow$ Breit–Wigner
D–T	ab initio	five nucleons, realistic forces

Lane & Thomas · Rev. Mod. Phys. 30, 257 (1958) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)
Navrátil & Quaglioni · Phys. Rev. Lett. 108, 042503 (2012)

p–p fusion from wave functions



$$H_0 = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}r^2}$$

deuteron $(H_0 + V_d)\,u_d = -B_d\,u_d$

p–p $(H_0 + V_{pp} + V_C)u_{pp} = E u_{pp}$

overlap $\Lambda \propto \int_0^\infty u_d\,u_{pp}\,\mathrm{d}r$

rate $S_{pp}(0) \propto G_V^2 g_A^2\, f_{pp}\,\Lambda^2$

V_d: S–D coupled by tensor force

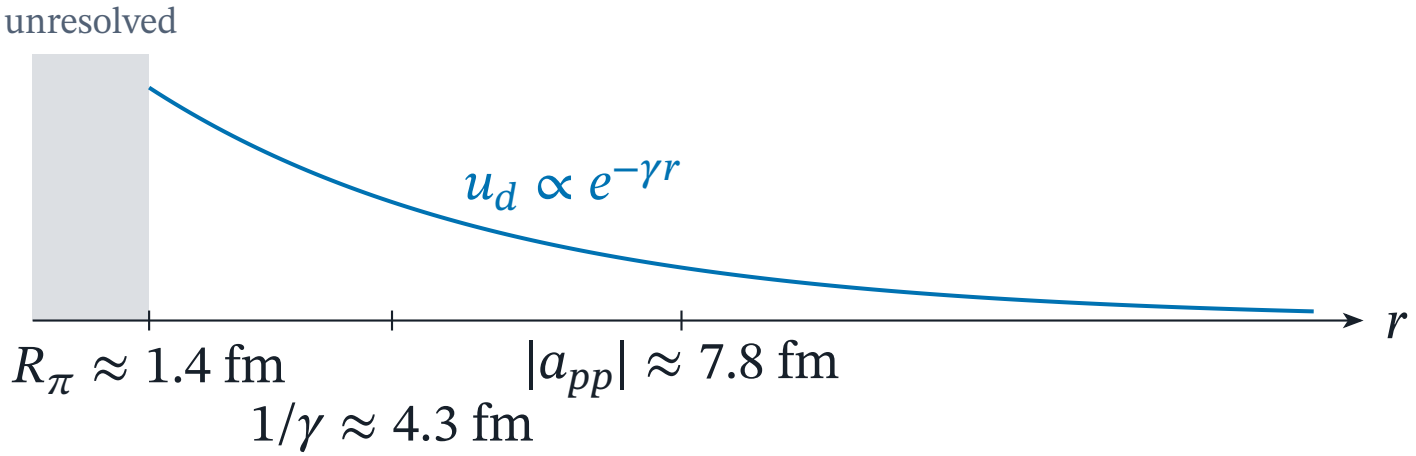
u_{pp}: $E \to 0$, matched to Coulomb waves, Gamow factor divided out

V: Reid potential, fitted to nucleon–nucleon scattering

f_{pp}: phase space of e^+ and ν

Reid · Ann. Phys. 50, 411 (1968) · Schiavilla et al. · Phys. Rev. C 58, 1263 (1998)

Pionless effective field theory



scale separation $R_\pi = \dfrac{\hbar}{m_\pi c} \ll \dfrac{1}{\gamma}, |a_{pp}|$

deuteron $\gamma = \dfrac{\sqrt{m_N B_d}}{\hbar}$

p–p $k \cot \delta = -\dfrac{1}{a_{pp}} + \dfrac{r_{pp}}{2} k^2 + \ldots$

short range one constant $L_{1,A}$ from lattice QCD

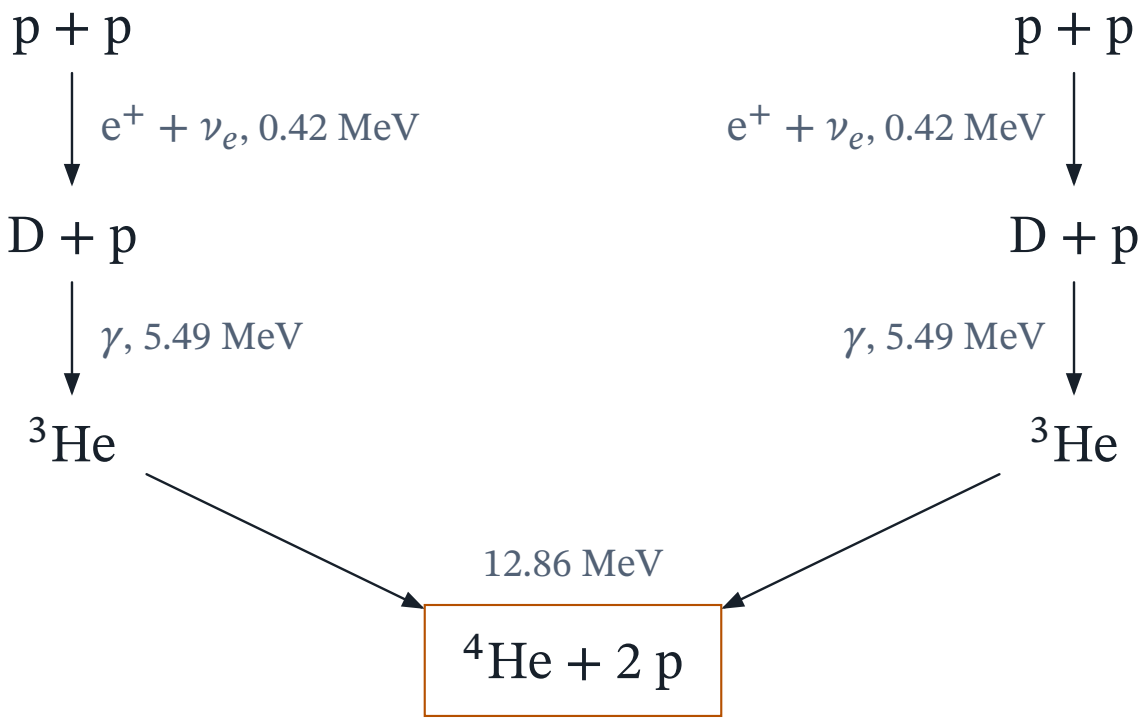
p–p: computed vs reference

	potential model	pionless EFT	reference
deuteron binding	2.2246 MeV	input	2.2246 MeV
D-state fraction	6.5%	–	$\sim 5-7\%$
a_{pp}	-7.75 fm	input	-7.81 fm
Λ^2	6.91	7.02	≈ 7.0
$S_{pp}(0)\,[10^{-25}\ \text{MeV b}]$	3.95	4.01 ± 0.27	4.01 ± 0.04

EFT order by order: $3.60 \rightarrow 3.82 \rightarrow 4.01$

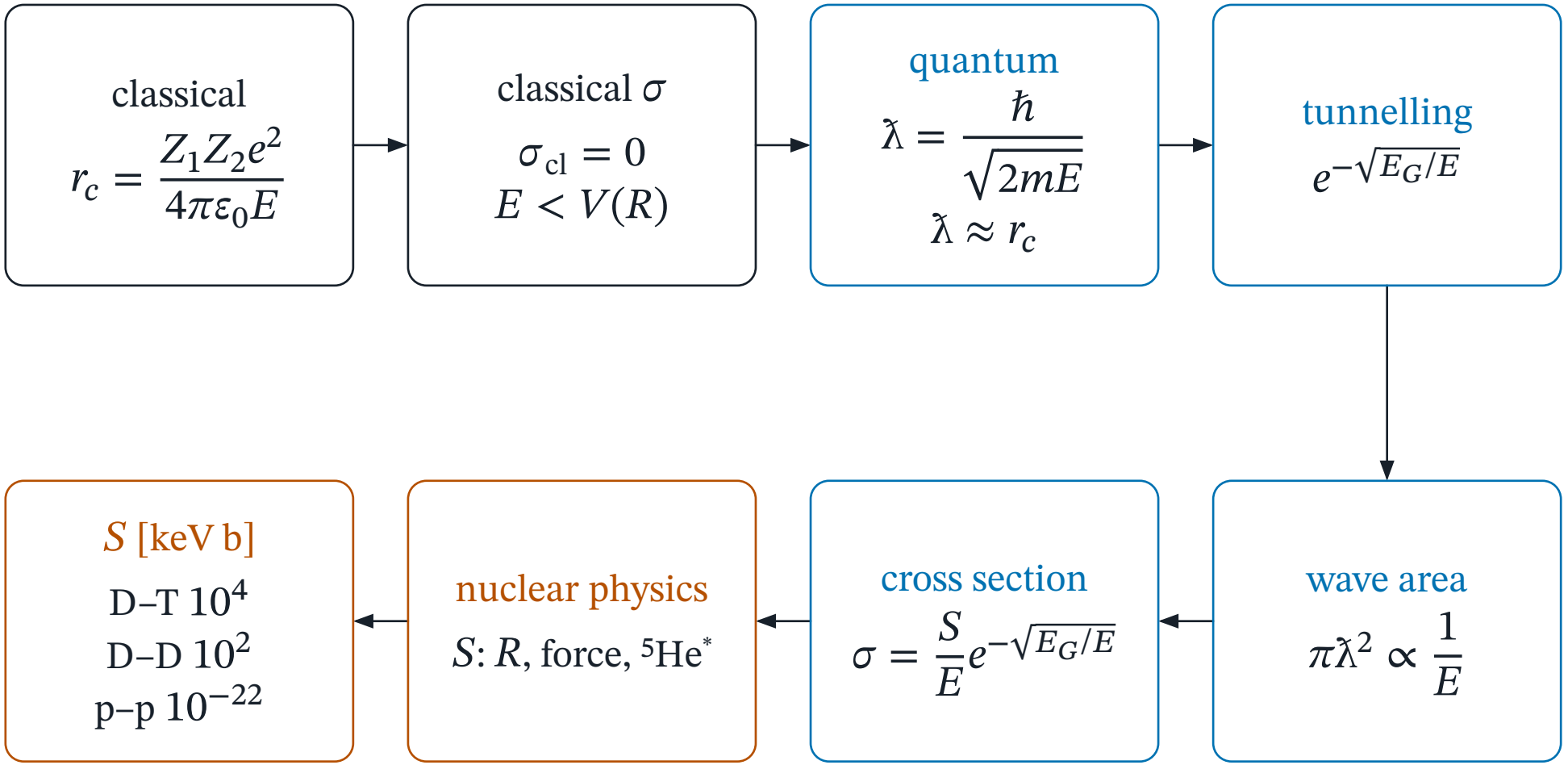
Butler & Chen · Phys. Lett. B 520, 87 (2001) · Savage et al. · Phys. Rev. Lett. 119, 062002 (2017)
Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)

Proton–proton chain

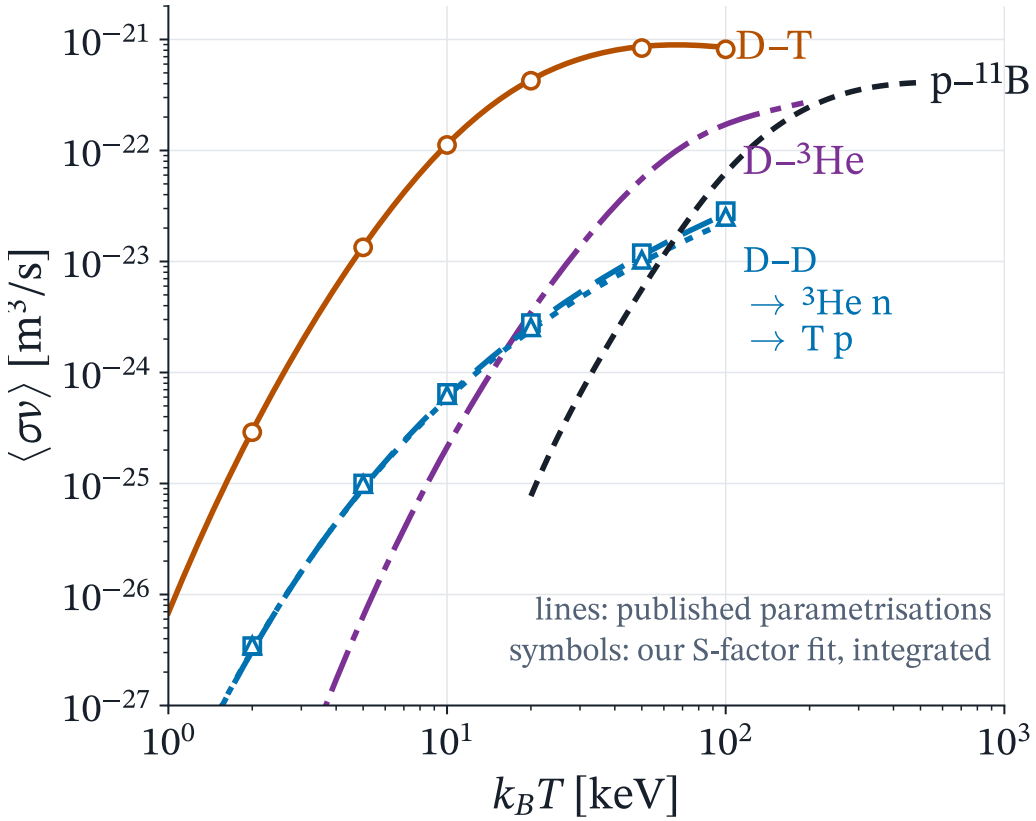


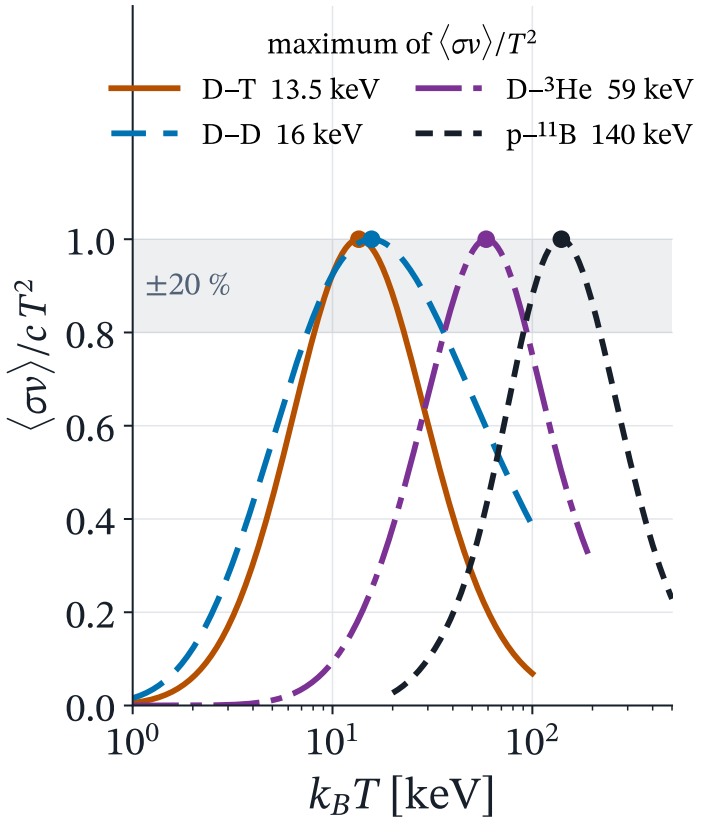
$4\,\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$ 26.7 MeV incl. $2 \times \mathrm{e}^+\mathrm{e}^-$ annihilation

From classical to nuclear



2.20 Reactivity





D–T, $k_B T = 10$ keV: $\langle \sigma v \rangle = 1.1 \times 10^{-22} \mathrm{m}^3/\mathrm{s}$

Lawson (Chapter 4): $nT\tau_E \propto T^2/\langle \sigma v \rangle$

Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Nevins & Swain · Nucl. Fusion 40, 865 (2000) · notebook: reactivity and Gamow peak

2.20 Reaction rate: summary

Assumptions

Maxwellians	both species at one T
$S(E)$	slowly varying near E_0

Derivation

rate	$R = n_1 n_2 \langle \sigma v \rangle / (1 + \delta_{12})$
relative motion	Maxwellian with m
average	$\langle \sigma v \rangle \propto (k_B T)^{-3/2} \int \sigma E e^{-E/k_B T}\, \mathrm{d}E$
in $\ln E$	T shifts a fixed kernel
Gamow peak	$E_0 = \left(E_G (k_B T)^2 / 4 \right)^{1/3}$

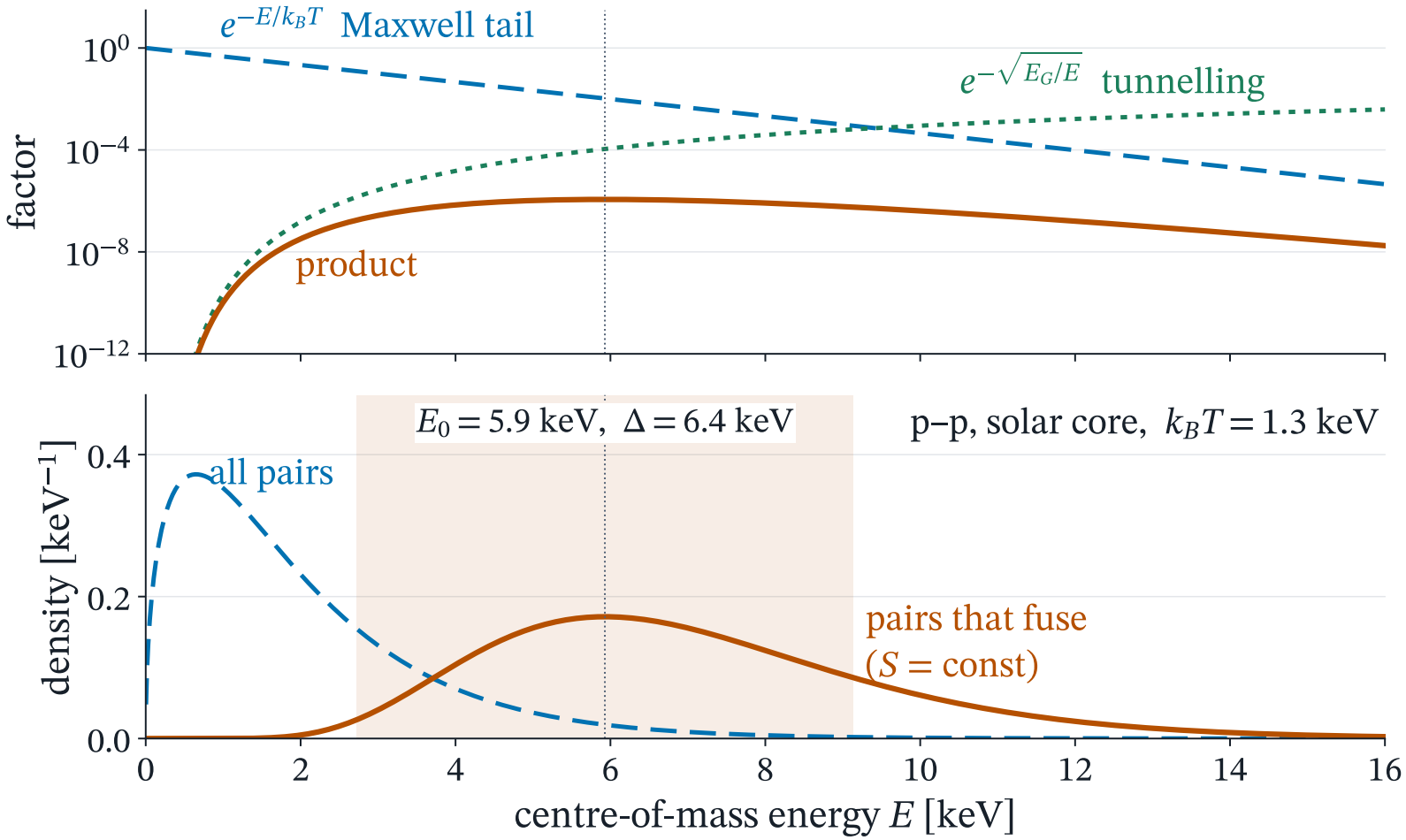
Result

$$\langle \sigma v \rangle \propto \tau^2 e^{-\tau}$$

$\tau = 3E_0 / k_B T$

D–T, $k_B T = 10$ keV:
$\langle \sigma v \rangle = 1.1 \times 10^{-22} \mathrm{m}^3/\mathrm{s}$
$\langle \sigma v \rangle / T^2$ maximal at 13.5 keV

2.20 Solar core



p–p: $E_G = 0.49$ MeV, $k_BT \approx 1.3$ keV $\Rightarrow$ $E_0 \approx 6$ keV $\approx 4.6\, k_BT$